C Security

Martin Johns

The C programming language

Developed 1972 by Dennis Ritchie

All major operating systems are written in C or C variants

- Windows
- Linux
- Mac Os X, Solaris, *BSD, ...

Most popular applications

- Microsoft Office
- Internet Explorer, Firefox
- Photoshop, iTunes, ...

Important Servers

- Web Server (Apache, IIS)
- Mail Server, DNS Server, SSH, Telnet, ...

The C programming language

C is a “low level” programming language

- Procedural
- No object-oriented features
- Pointers

C is compiled into machine/binary code

No memory abstraction

- All variables in C programs correspond directly to specific location in the RAM

C successors

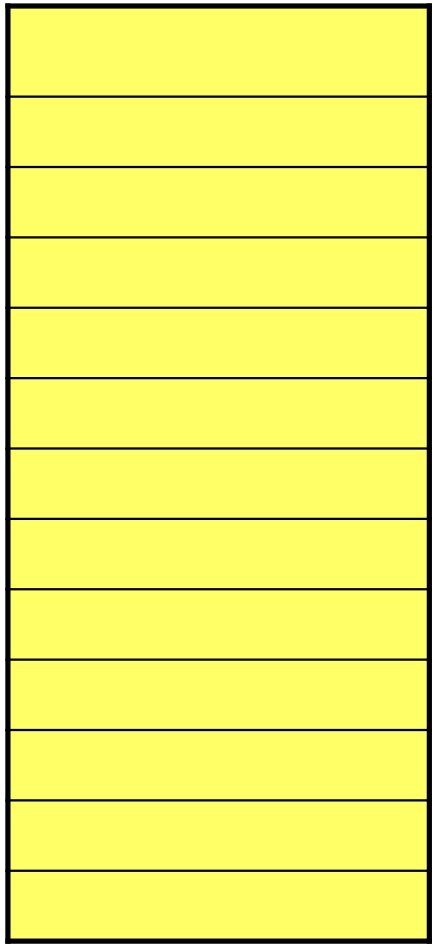
C++

- Developed 1979 von Bjarne Stroustrup
- Superset of C
- Object-oriented
- Tons of advanced/complex features
  - Templates, operator overloading, polymorphism, multiple inheritance
  - But no memory safety
- In heavy use at Microsoft

Objective C

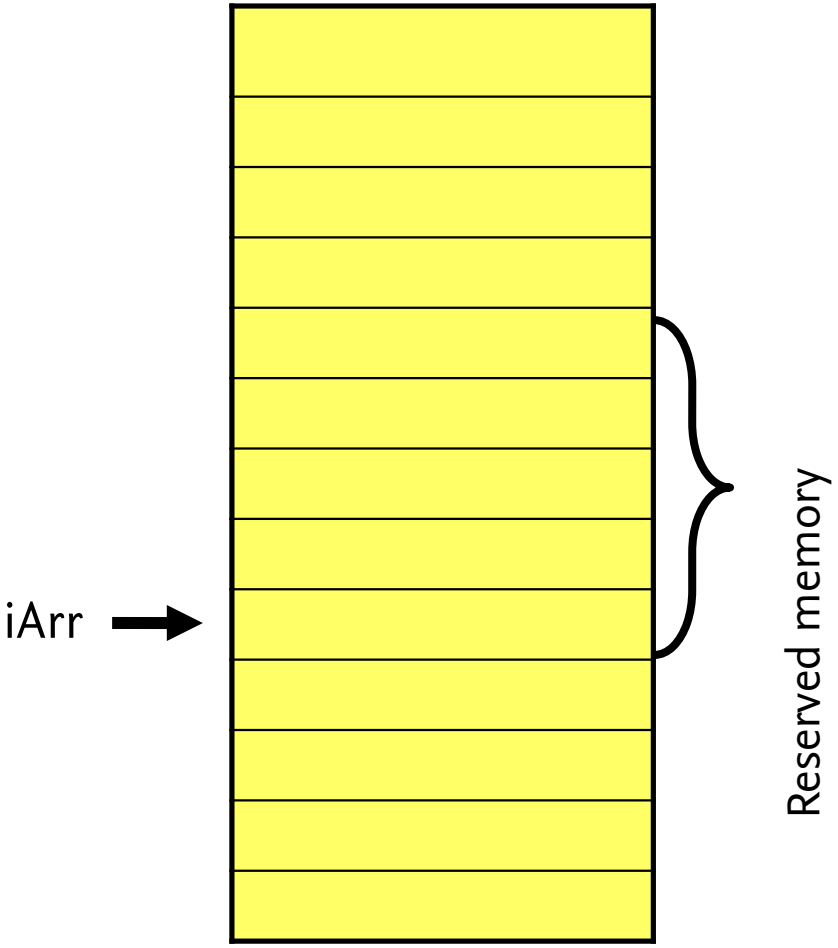
- Also an object-oriented superset of C
- Small talk like message passing
- Dynamic typing
- But no memory safety
- In heavy use at Apple / iOS

Memory management



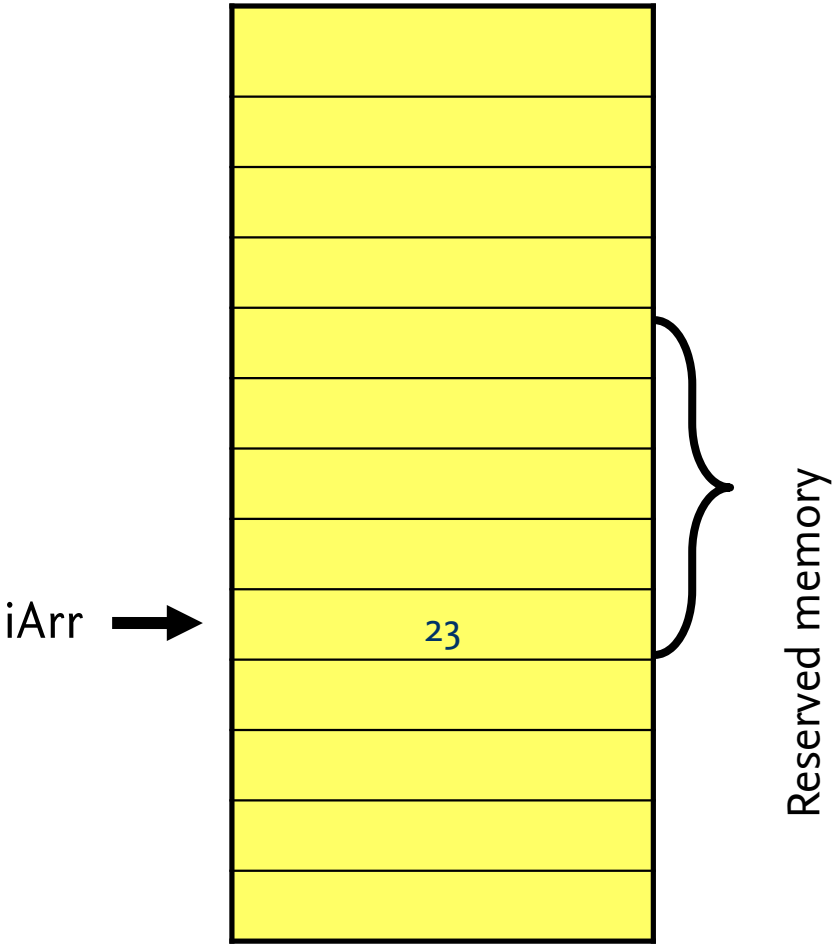
```
int iArr [5];
```

Memory management



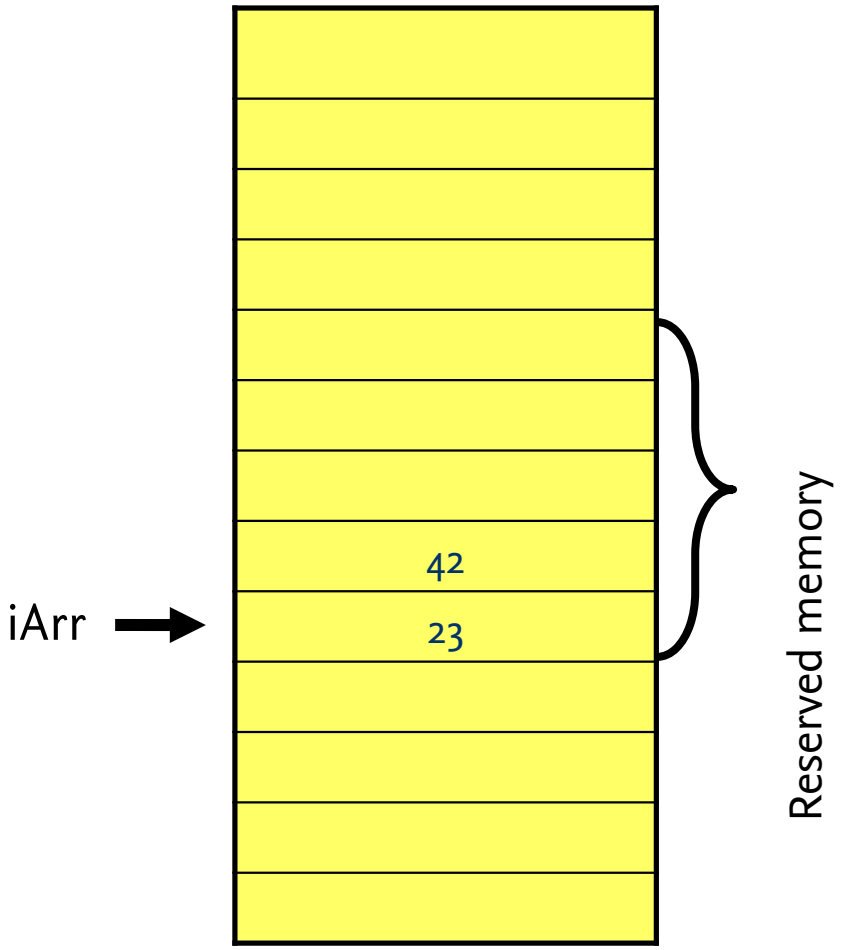

```
int iArr [5];
iArr[0] = 23;
```

Memory management



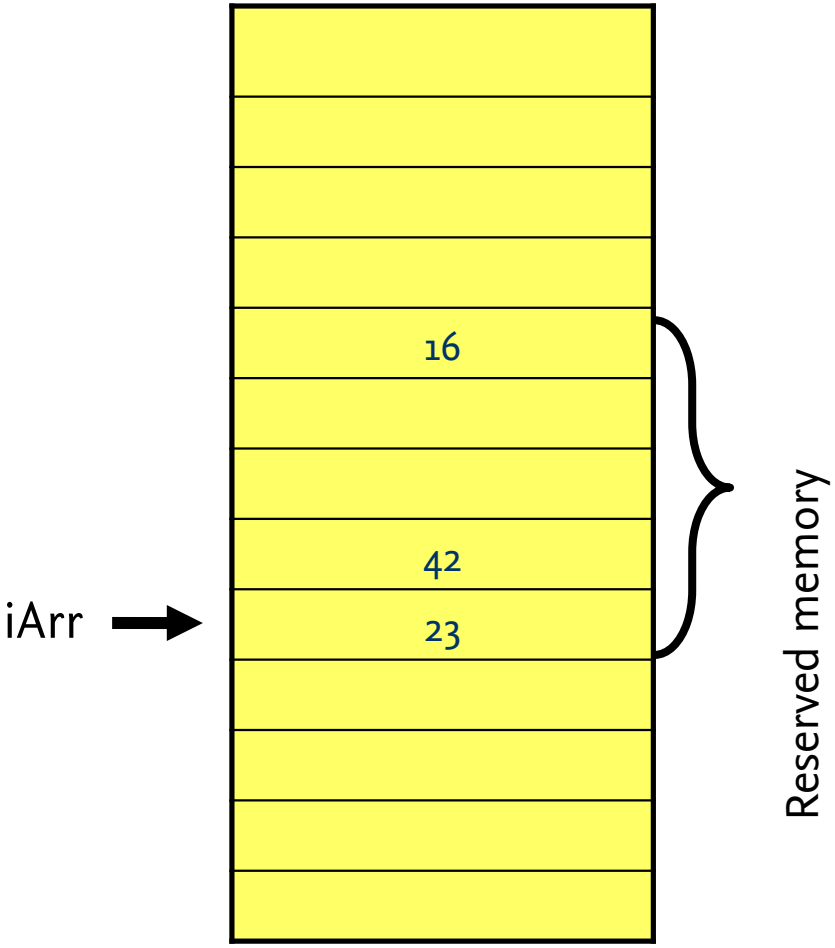

```
int iArr [5];
iArr[0] = 23;
iArr[1] = 42;
```

Memory management




```
int iArr [5];
iArr[0] = 23;
iArr[1] = 42;
iArr[4] = 16;
```

Memory management



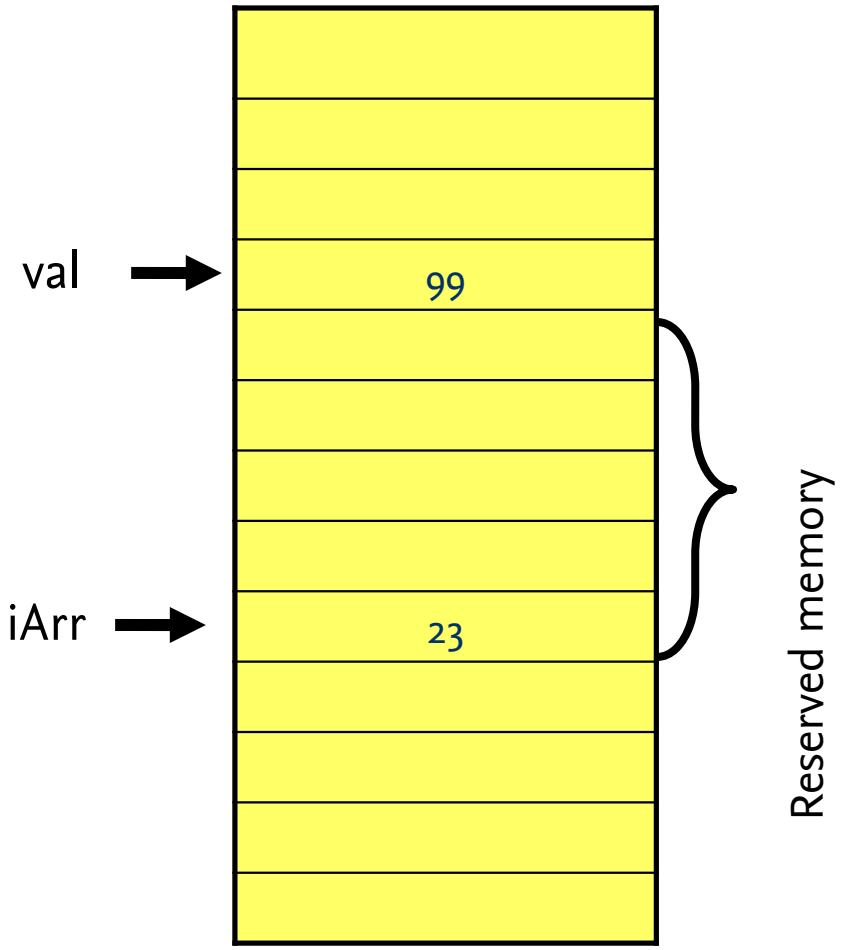

```
int iArr [5];
iArr[0] = 23;
iArr[1] = 42;
iArr[4] = 16;
```

Buffer Overflow

The buffer boundaries are not enforced!

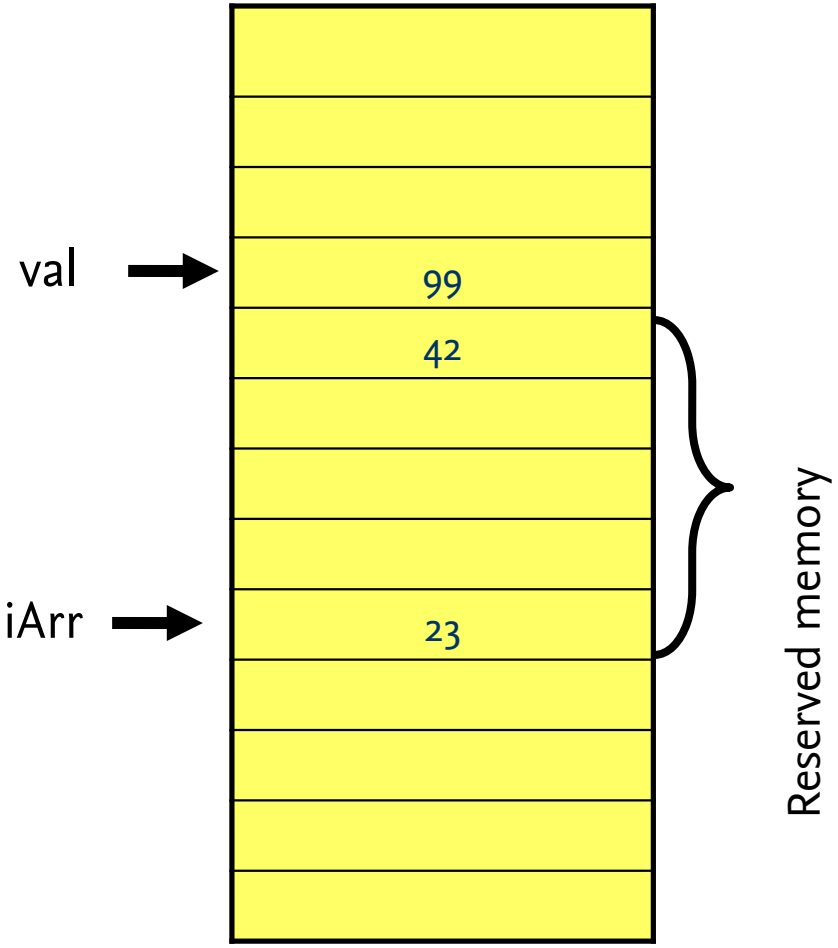
what happens if a misbehaving program violates them?

Memory management



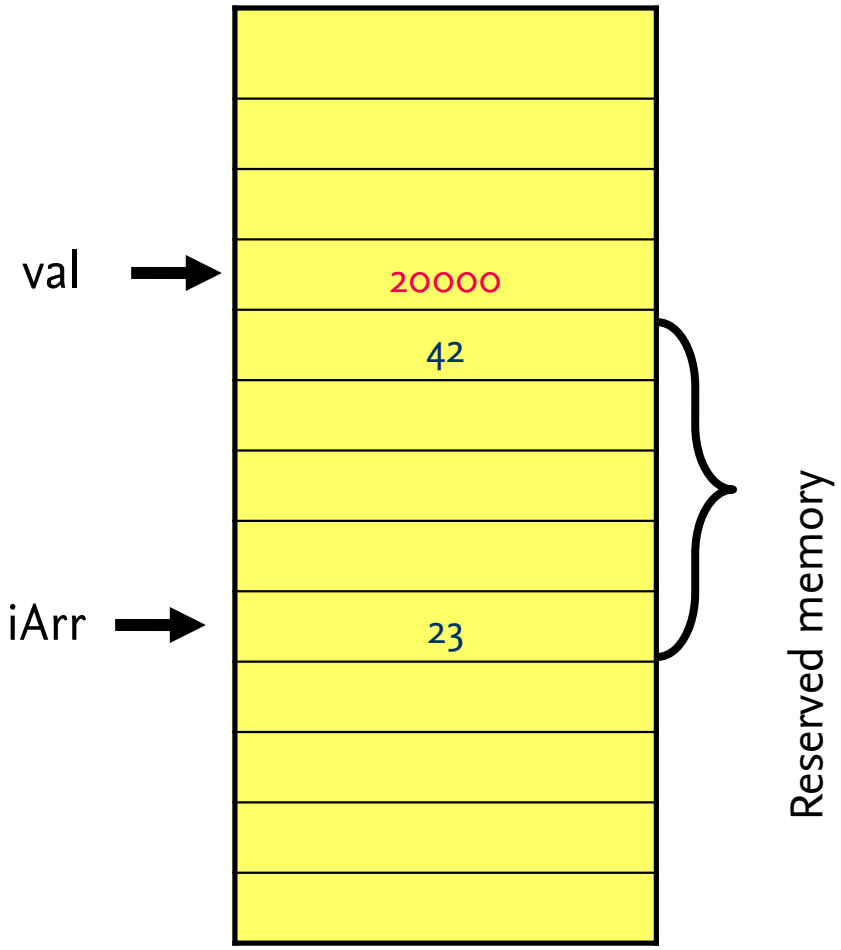

```
int val = 99;
int iArr [5];
iArr[0] = 23;
iArr[4] = 42;
```

Memory management




```
int val = 99;
int iArr [5];
iArr[0] = 23;
iArr[4] = 42;
iArr[5] = 20000;
```

Memory management




```
int val = 99;
int iArr [5];
iArr[0] = 23;
iArr[4] = 42;
iArr[5] = 20000;
```

Machine Code

Executed directly by the processor

Simple instruction format:

Do *A* with *B* and store the result in *C*

- Assignments
- Jumps
- Conditionals

No variables / types

- Instead: Memory locations
  - Registers
  - RAM
- Memory locations are addressed
  - Absolute
  - Relative
  - In both cases some absolute values are needed

Subroutines

A function in the C code is translated to a subroutine in machine code

Instructions

- Jump to subroutine at address *X*
- return

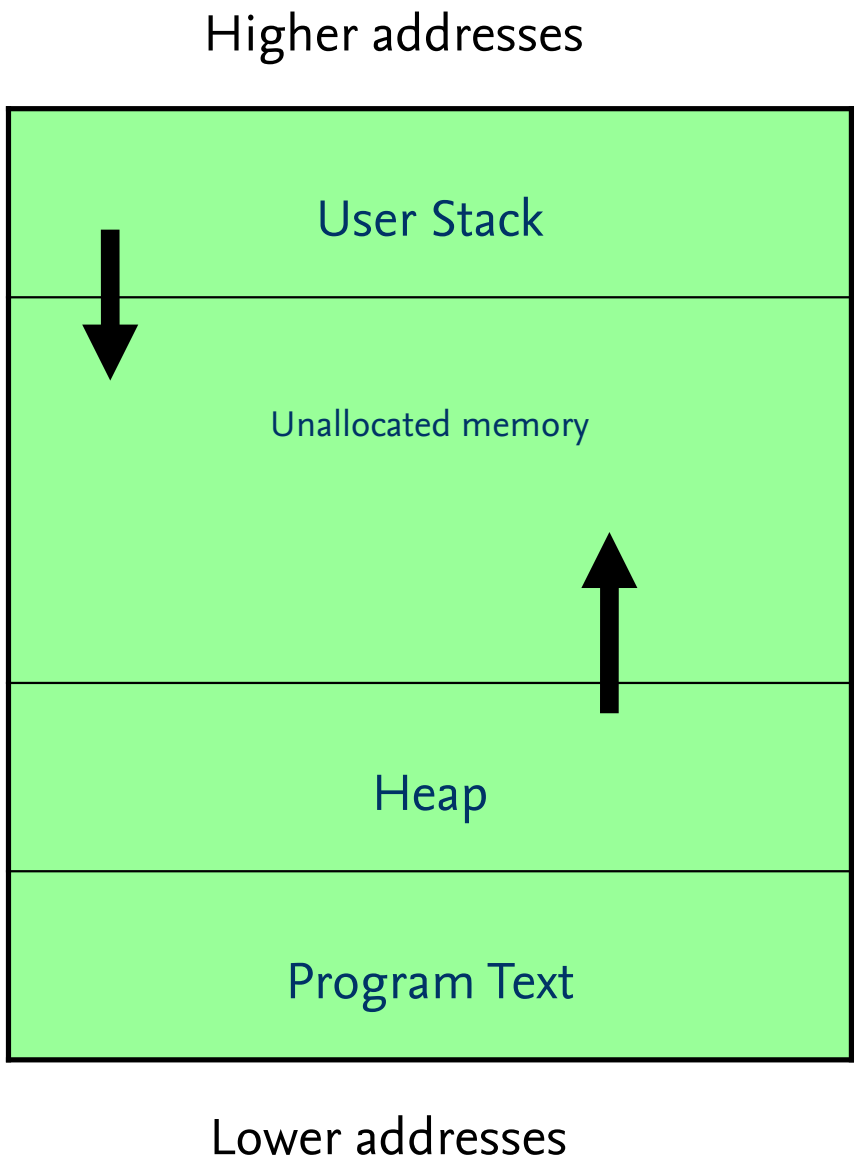
At compilation-time it is unknown how often a given subroutine is called

- How much memory should be reserved?
- How are the local values addressed?
- Where should the execution return after the subroutine has finished?

The Stack

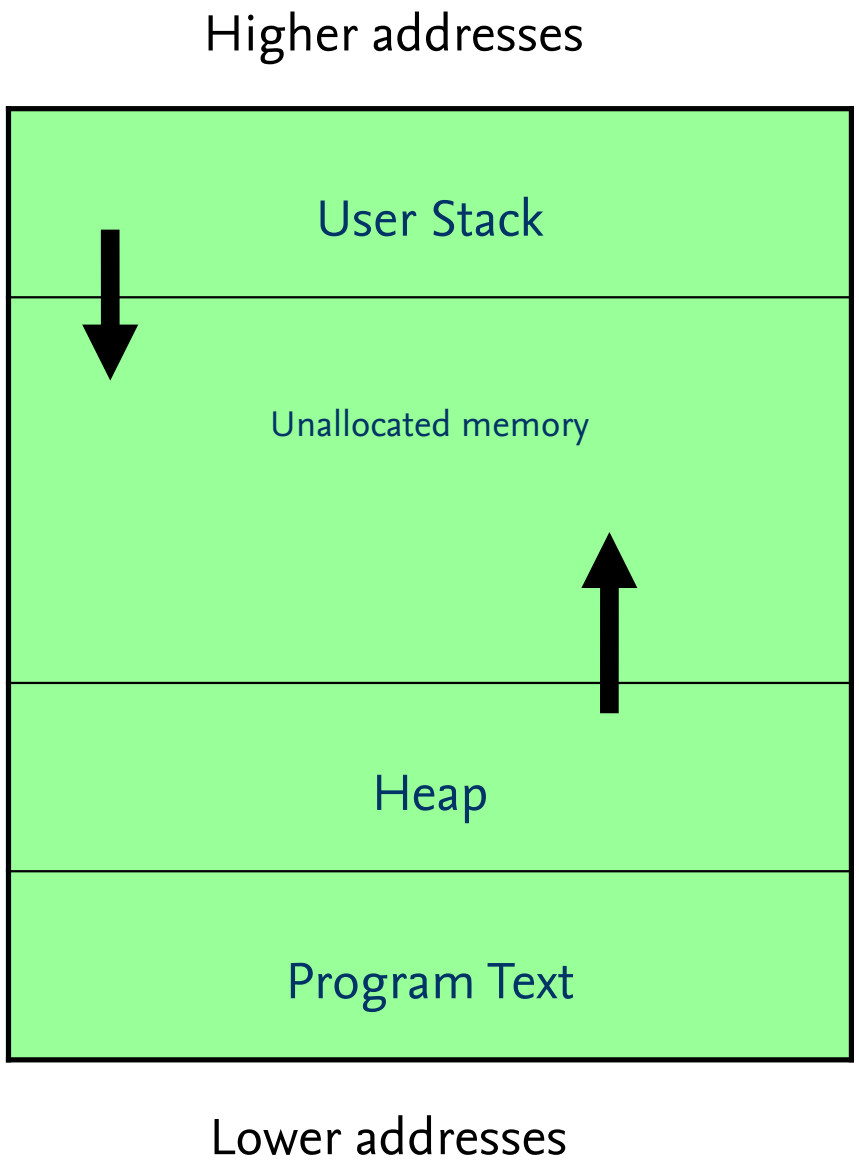
- A dedicated memory area
- Holds dynamically data which is created
  - Local variables
  - Execution-context information

Basics: Process Memory Layout



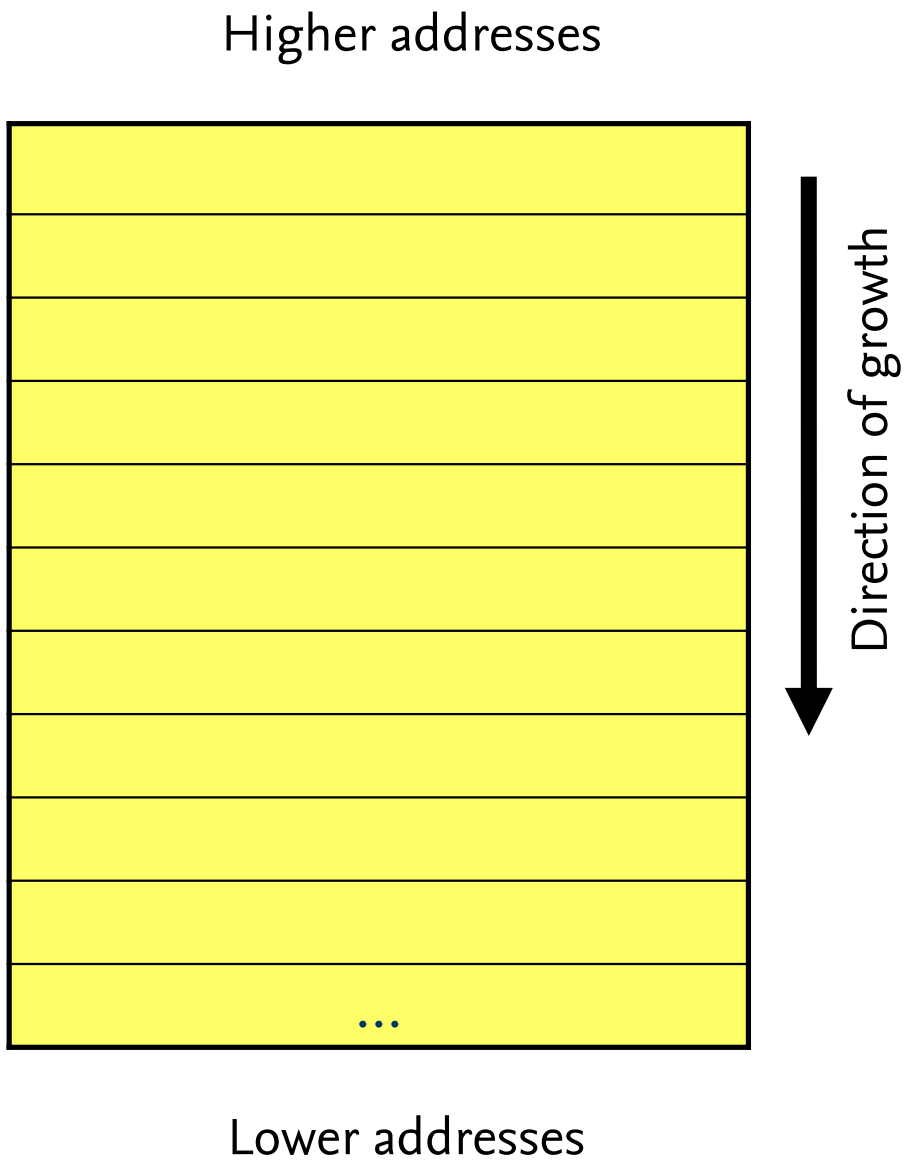
- Depending on the operating system the actual memory layout may somewhat differ
  - Usage of memory below the heap
  - Usage of memory above the stack
- Coherent memory is always read and written from lower addresses to higher addresses (not considering little/big endian)

Basics: The Stack vs. The Heap

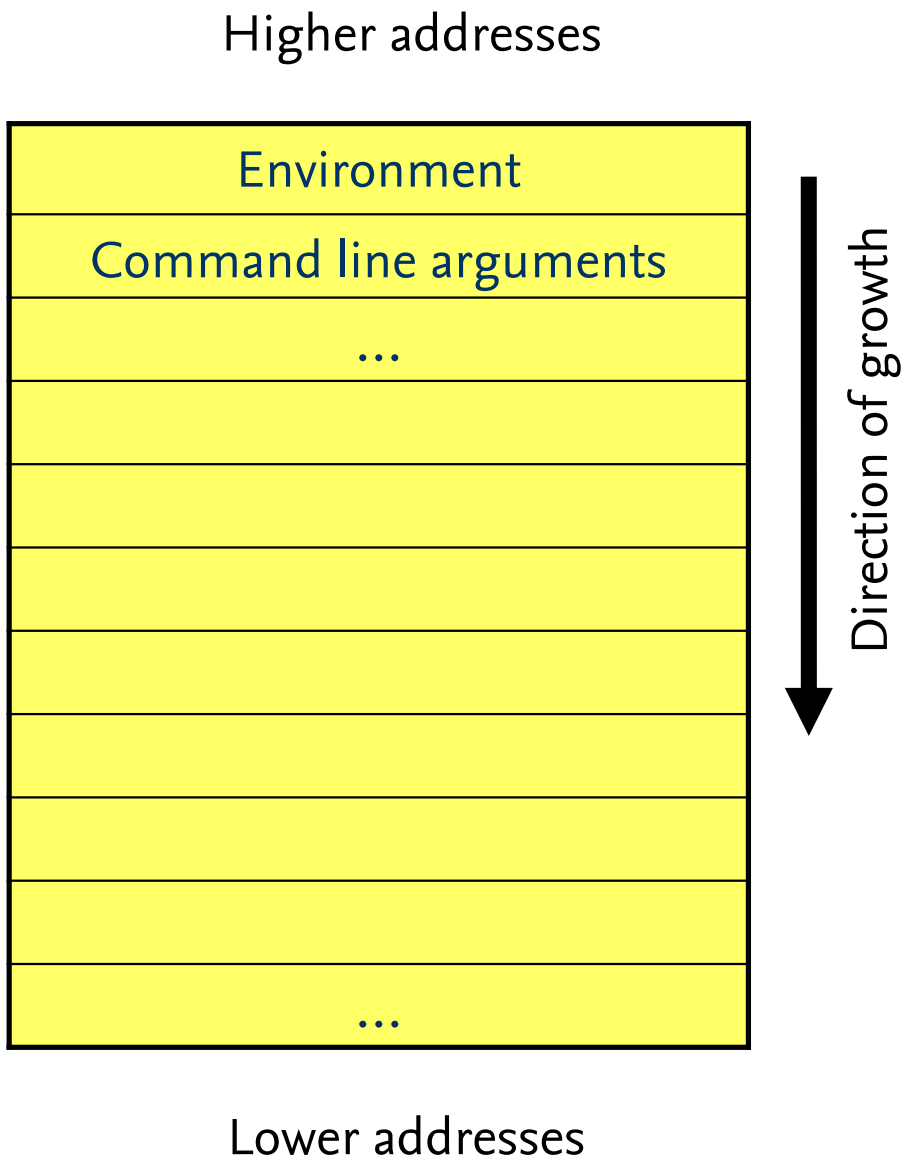


- The Stack grows from the higher to the lower addresses
- The Heap (initially) grows from the lower to the higher addresses
- The Stack is used to store data that is local to functions
- The Heap is used to store data that is persistent between function calls
- Stack allocation is implicit
  `char test[42];`
- Heap allocation is (usually) explicit
  `char *test = malloc(42*sizeof(char));`

Basics: The Stack



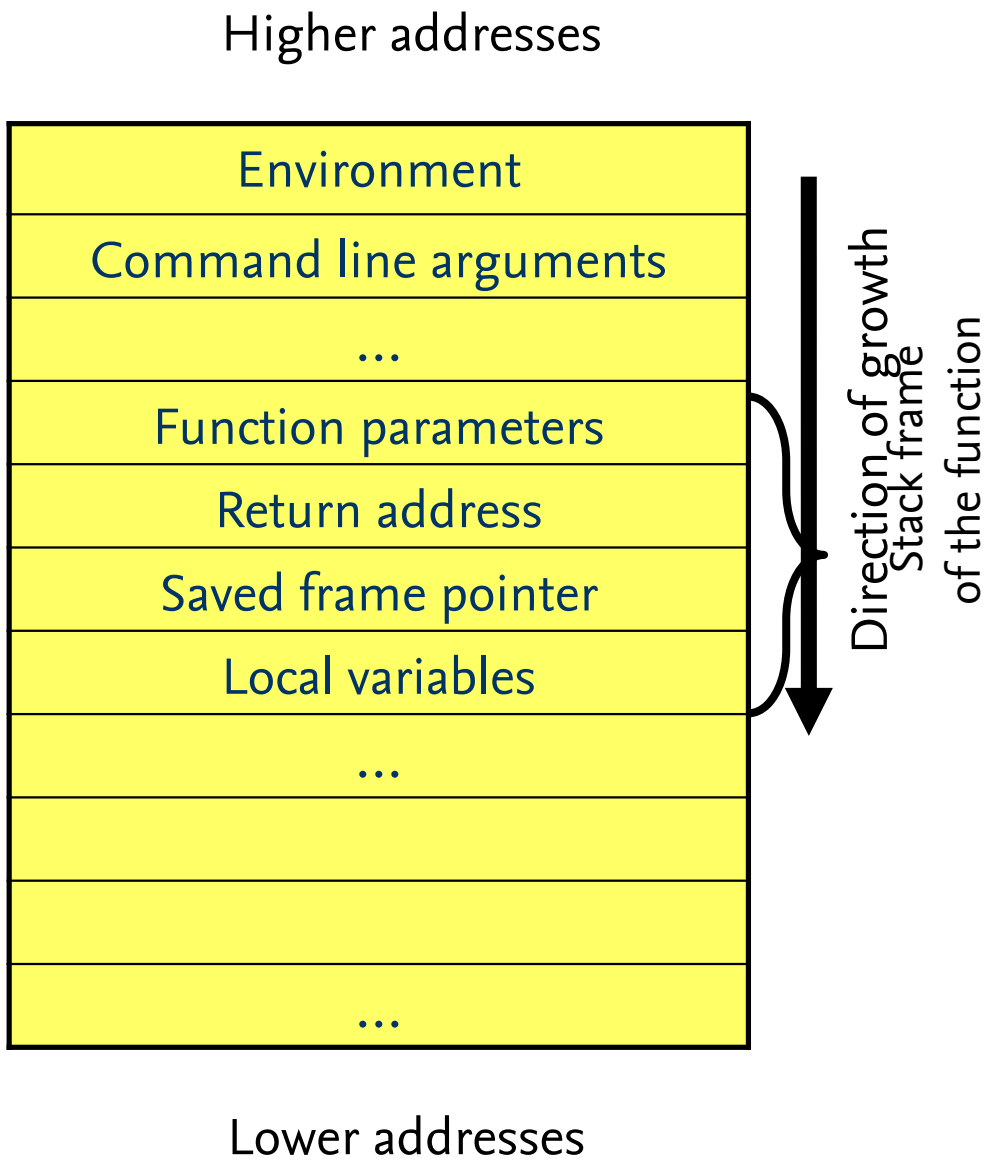
Basics: The Stack



The Stack is used for:

- The Environment
- Command line arguments

Basics: The Stack



The Stack is used for:

- The Environment
- Command line arguments
- Function parameters
- Return addresses
- Saved frame pointers
- Local variables

Basics: The Stack – Processor view (x86)

Several processor inherent constructs define the Stack

- The register SP points to the top of the stack
  - Instructions that are stack-aware modify the value of SP automatically
    - push, pop
    - call, return
    - (enter, leave)

The Frame Pointer

- The Frame Pointer is utilized by a function to access the local function context on the stack (function parameters, local variables)
- The register BP is used as Frame Pointer (assigned by the compiler)
  - Instructions to modify this register are (originally) generated by the compiler
  - Local variables and function parameters are referenced relatively to the Frame Pointer
    - Function parameters have positive offsets
    - Local variable have negative offsets

The stack

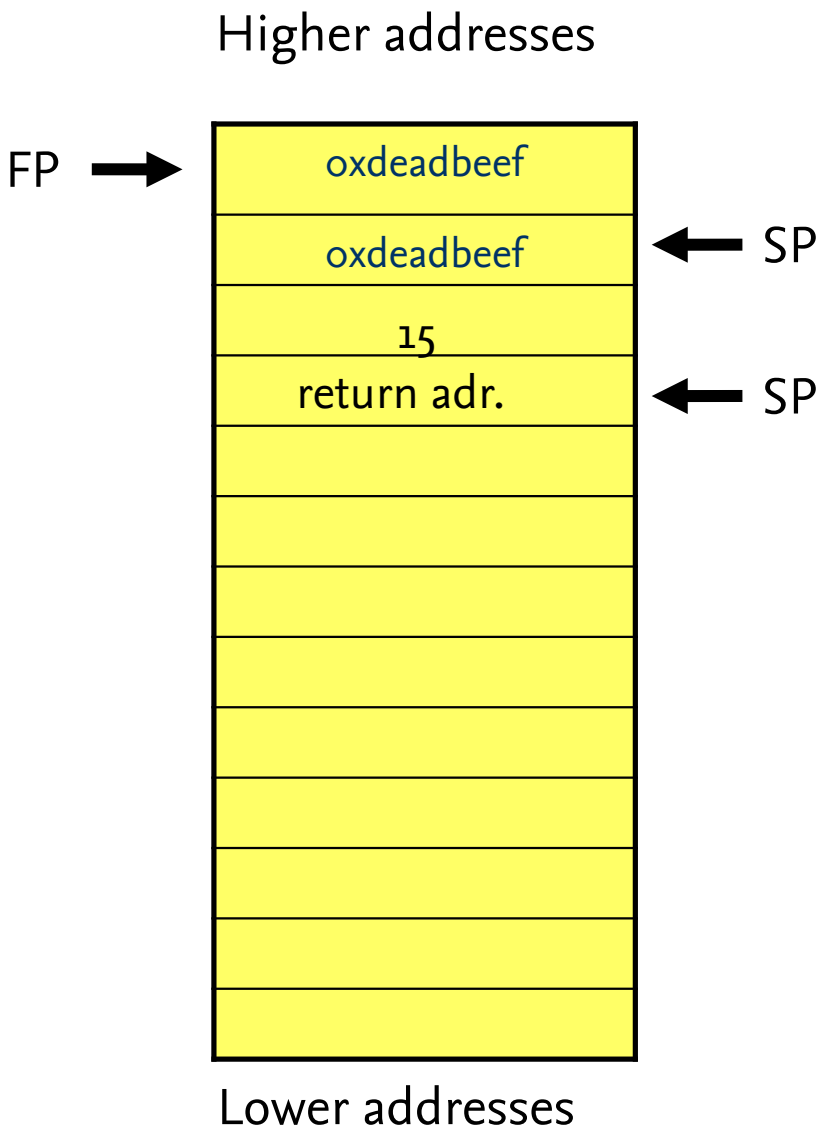
Instructions:

- push(val): Put val on top of the stack
- pop: Get the value that is on top of ths stack

Important registers:

- IP: Points to the next instruction
- SP: Points to the top if the stack
- FP: Defines the local data-context of a function on the stack
  - All local memory-addresses of the function are relative to FP

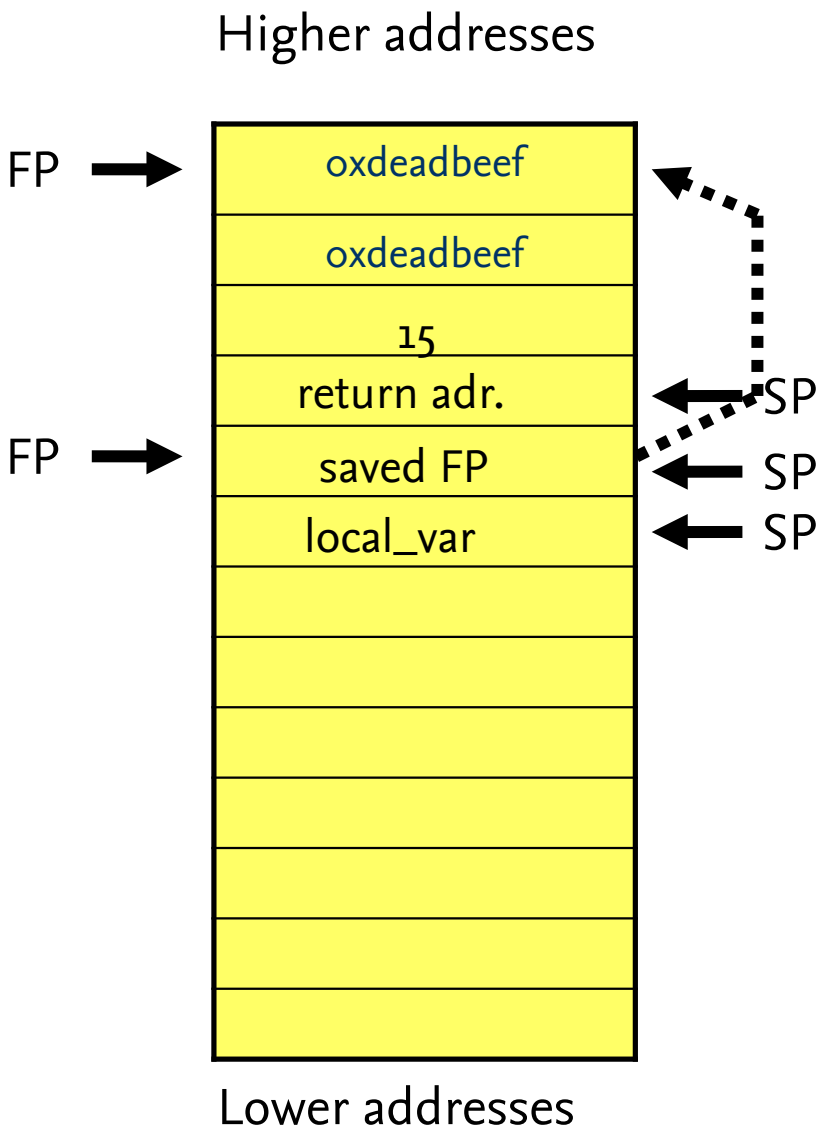
Basics: The function call




```
b = something(15);
```

- Move SP to reserve space for function parameter
- Write value of function parameter on the stack
- Call function (implicitly moves return address on the stack)

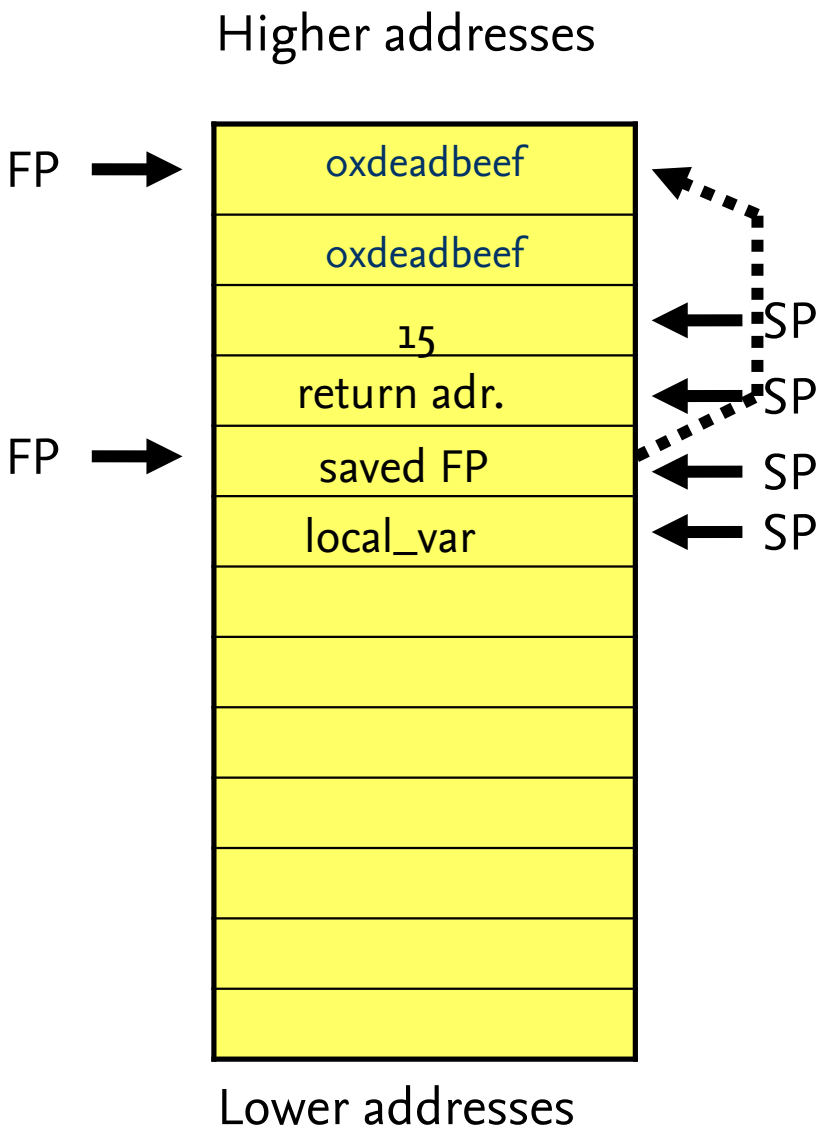
Basics: The function prolog




```
int something(int para){
   int local_var;
   ...
}
```

- Save the old FP on the stack
- Move the FP to the new frame
- Move the SP to reserve space for the local variables

Basics: The function epilog

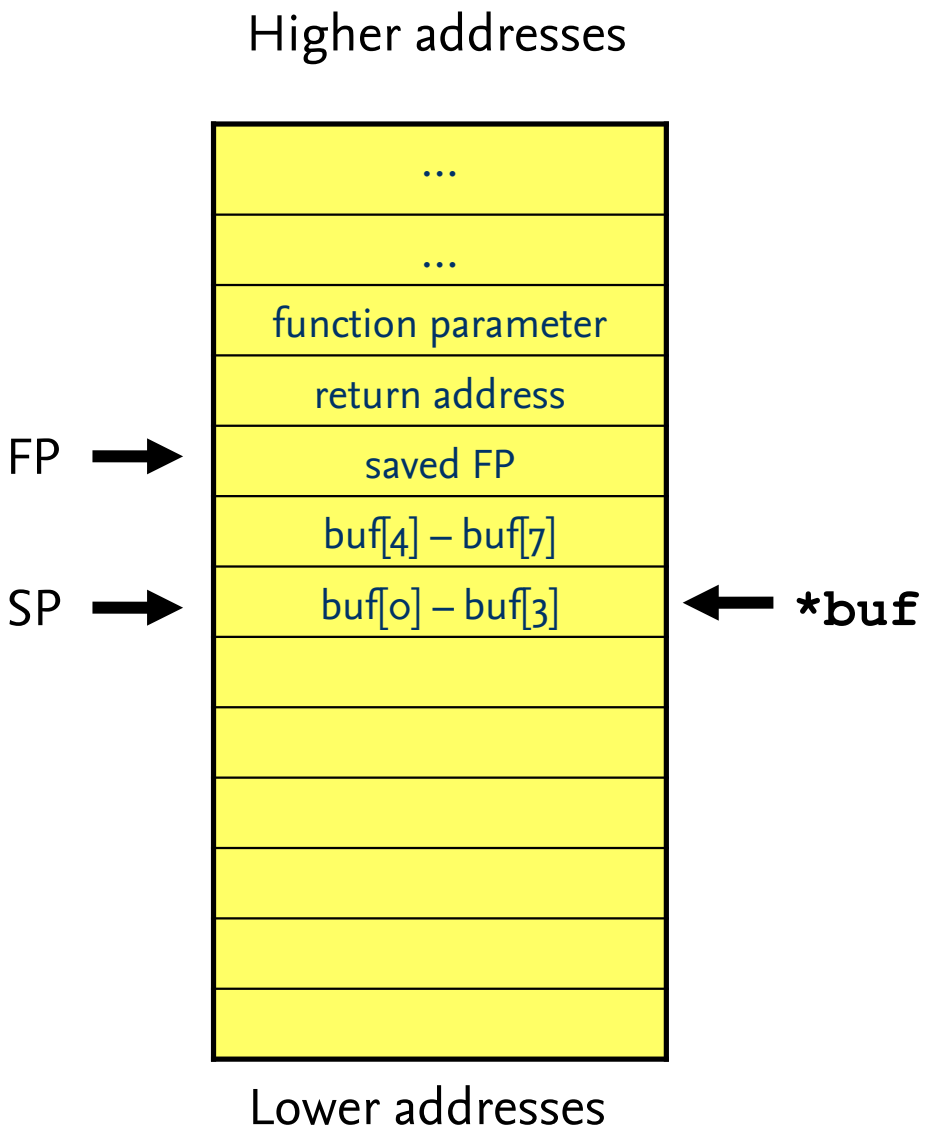



```
int something(int para){
   ...
   return (0);
}
```

- Write the return value to the register AX
- Move the SP back to the FP
- Restore the old FP
- Pop the stored old FP
- Pop the return address and leave the function

Buffer Overflows on the stack

Buffer Overflow on the stack

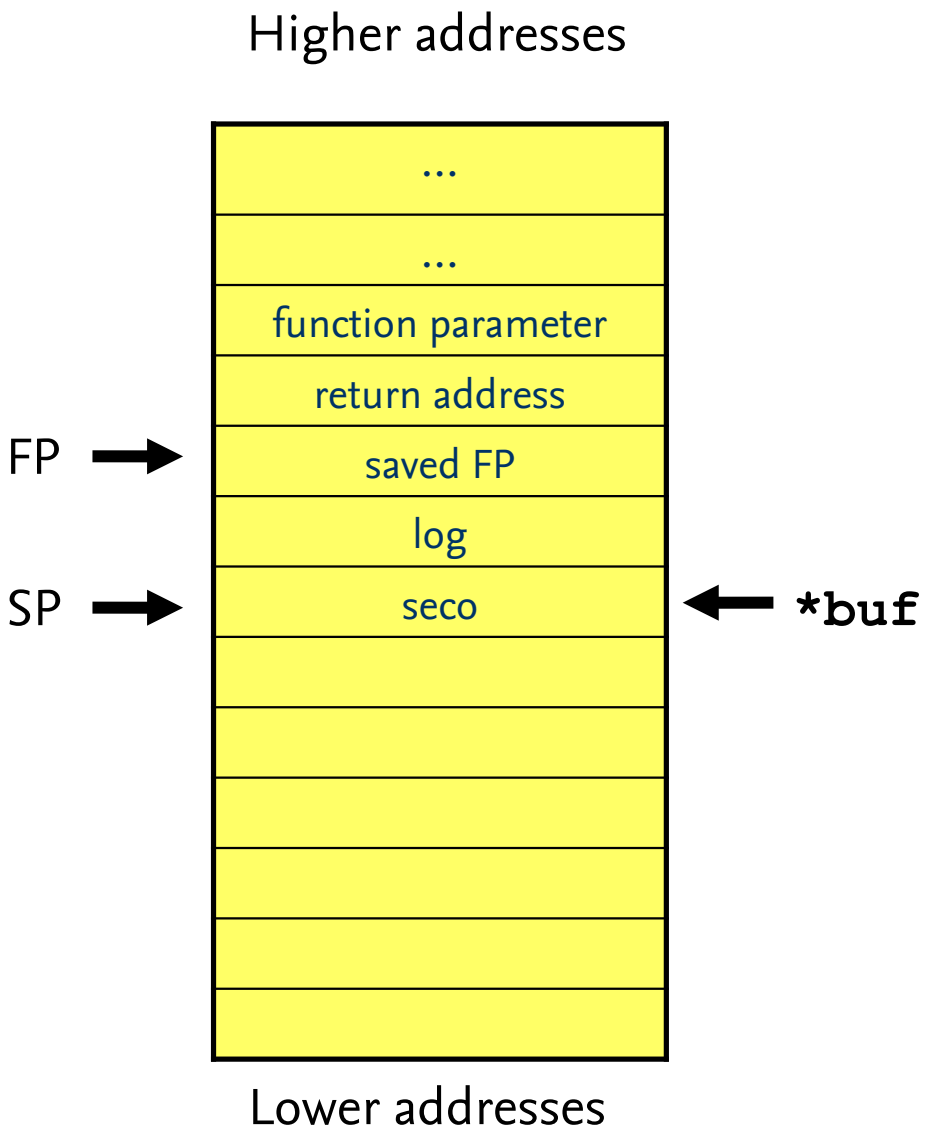



```
int something(int para){
  char buf[8];
  ...
  strcpy(buf, "secolog");



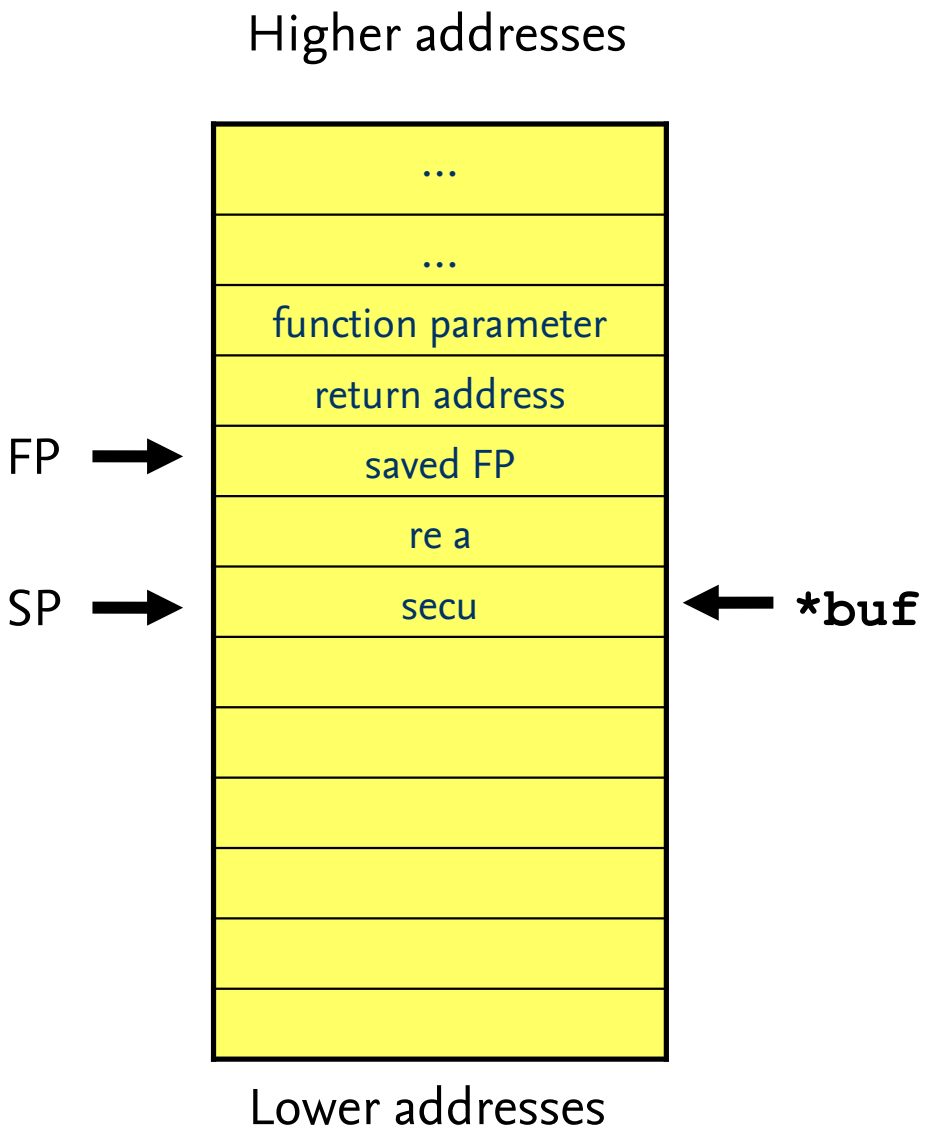
}
```

Buffer Overflow on the stack




```
int something(int para){
  char buf[8];
  ...
  strcpy(buf,
  "secolog"); ...
  strcpy(buf,
         "secure
  analysis");
}
```

Buffer Overflow on the stack




```
int something(int para){
  char buf[8];
  ...
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  strcpy(buf,
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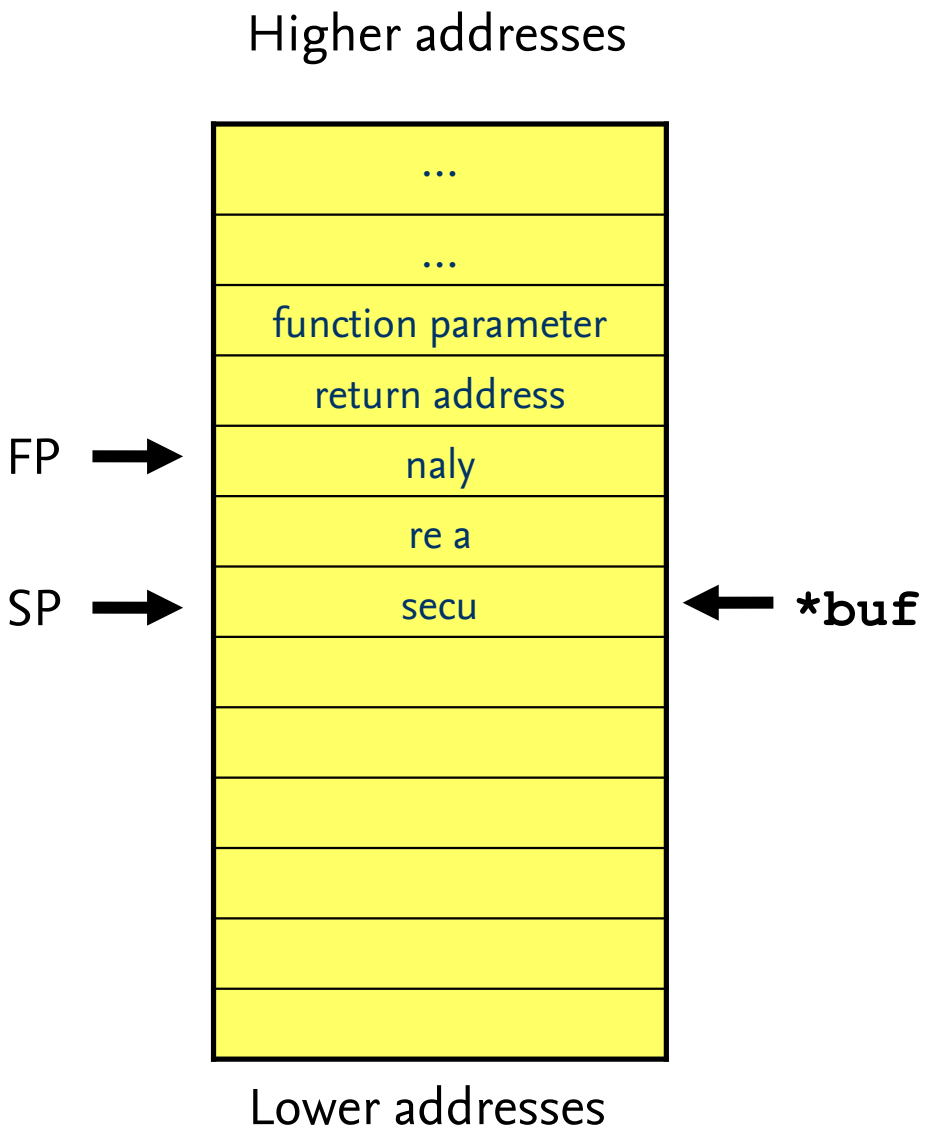
}
```

- The saved Frame Pointer gets overwritten

Buffer Overflow on the stack

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  ...
  strcpy(buf,
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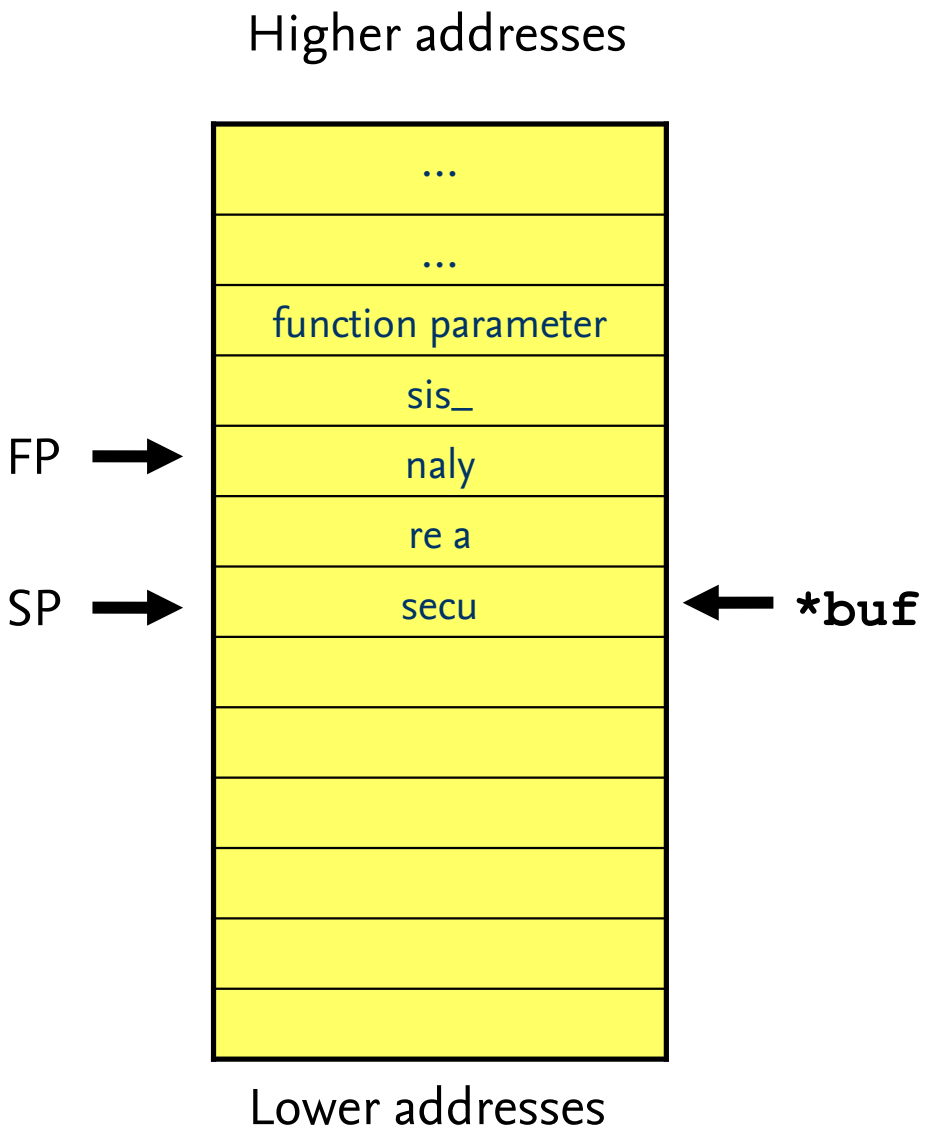
}
```

- The saved Frame Pointer gets overwritten
- The return address gets overwritten

Buffer Overflow on the stack




```
int something(int para){
  char buf[8];
  ...
  strcpy(buf,
  "secolog"); ...
  strcpy(buf,
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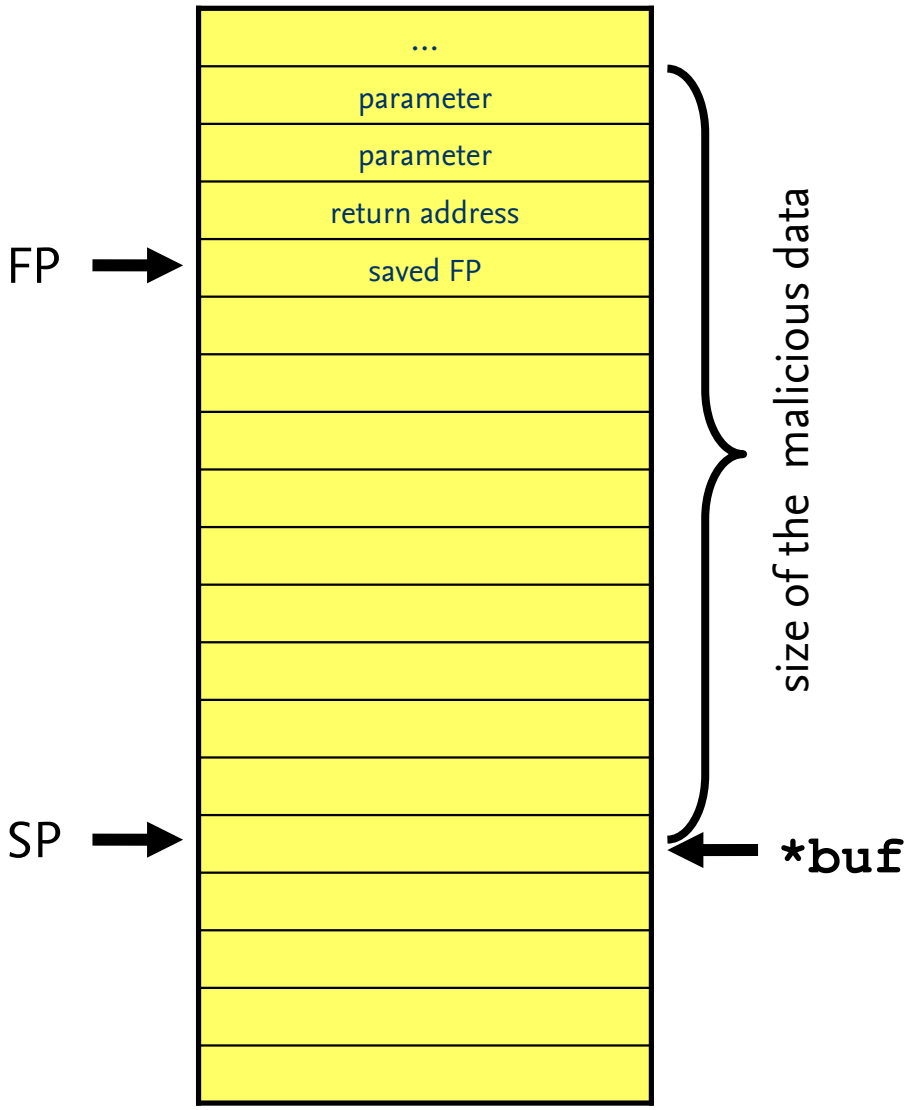
  analysis");
}
```

- The saved Frame Pointer gets overwritten
- The return address gets overwritten

Using a stack based Buffer Overflow to hijack execution

- Construct a buffer that contains executable code (the shellcode)
- Guess the memory address of the vulnerable buffer on the stack
- Pad the constructed shellcode
  - Before the shellcode: As many NOP instructions as possible
  - Behind the shellcode: The guessed address of the beginning of the shellcode (or the nops)
  - The constructed buffer has to be big enough to overwrite the return address on the stack

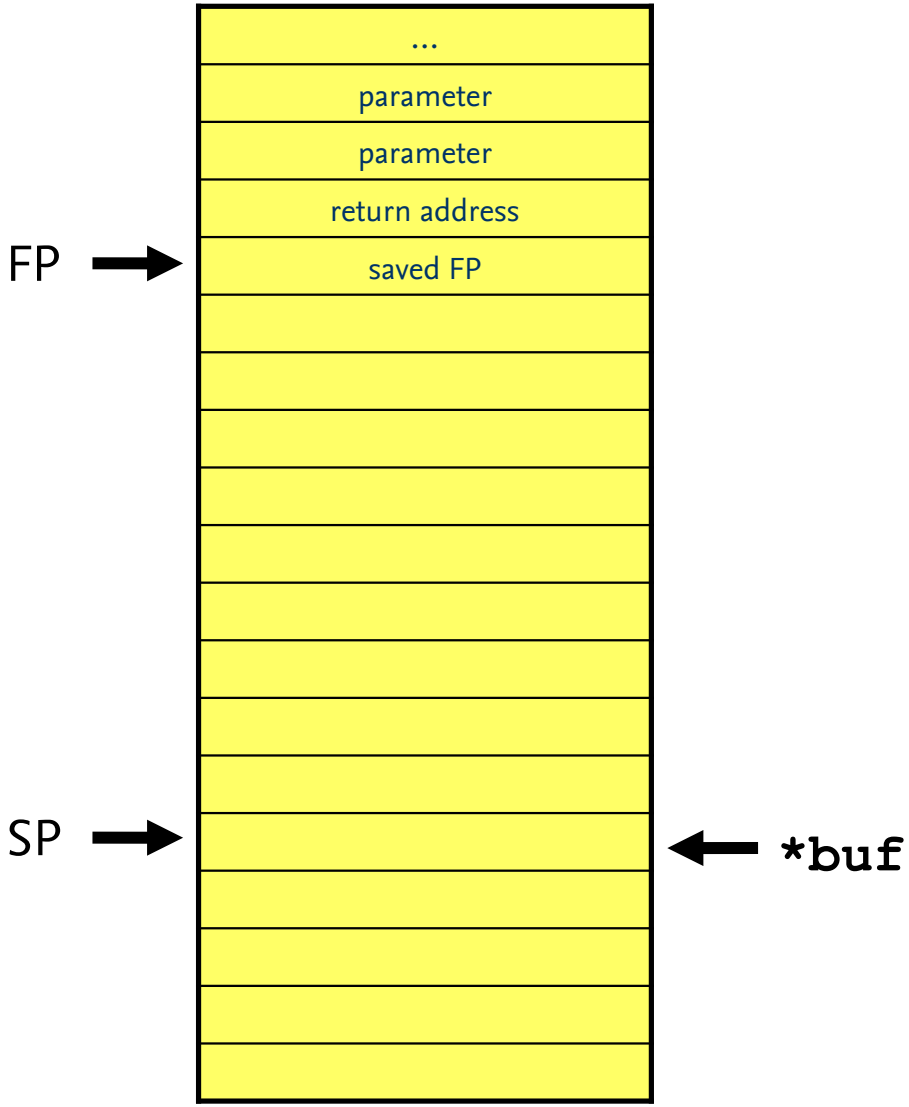
Buffer Overflow on the stack




```
int something(int,int){
   char buf[512];
   ...
   strcpy(buf,argv[1]);
}
```

- argv[1] contains the constructed malicious data
- The malicious data is big enough to overflow the buffer and overwrite the rerturn address

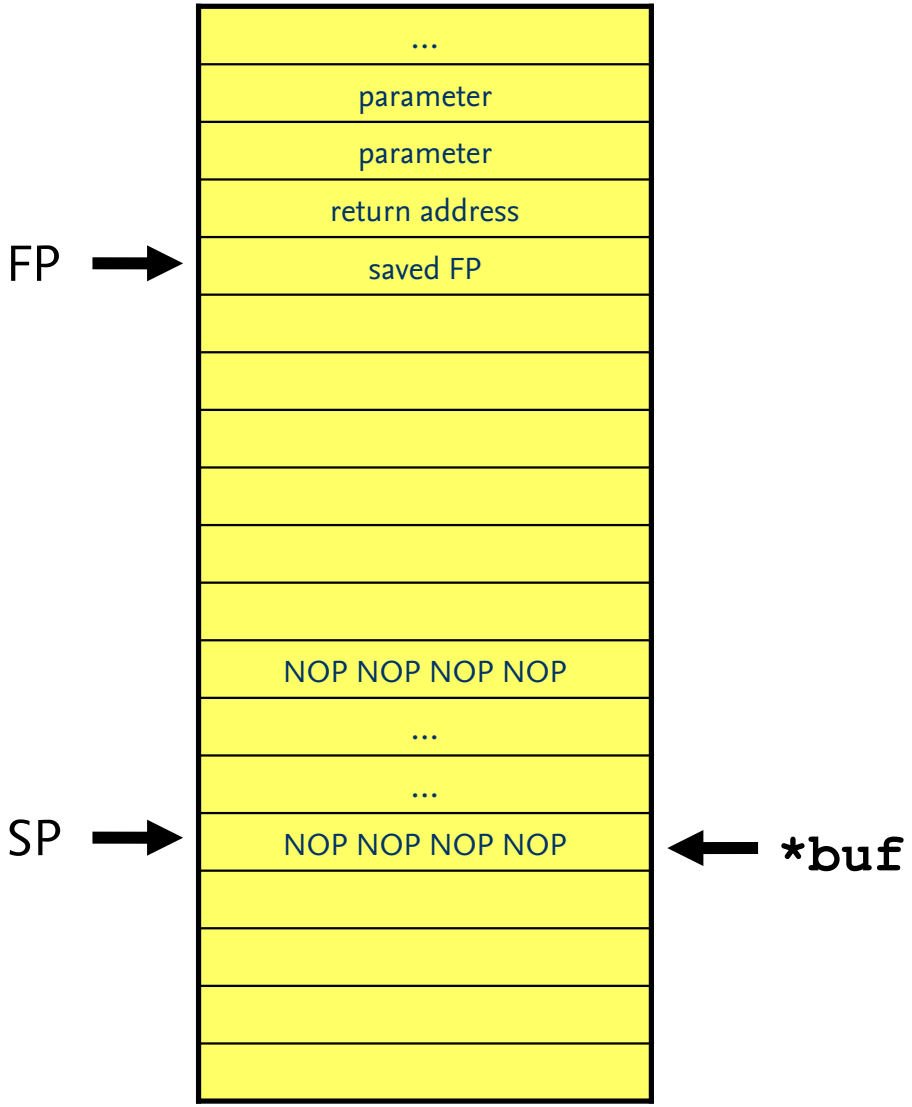
Buffer Overflow on the stack




```
int something(int,int){
   char buf[512];
   ...
   strcpy(buf,argv[1]);
}
```

- argv[1] contains the constructed malicious data
- The data starts with the NOPs

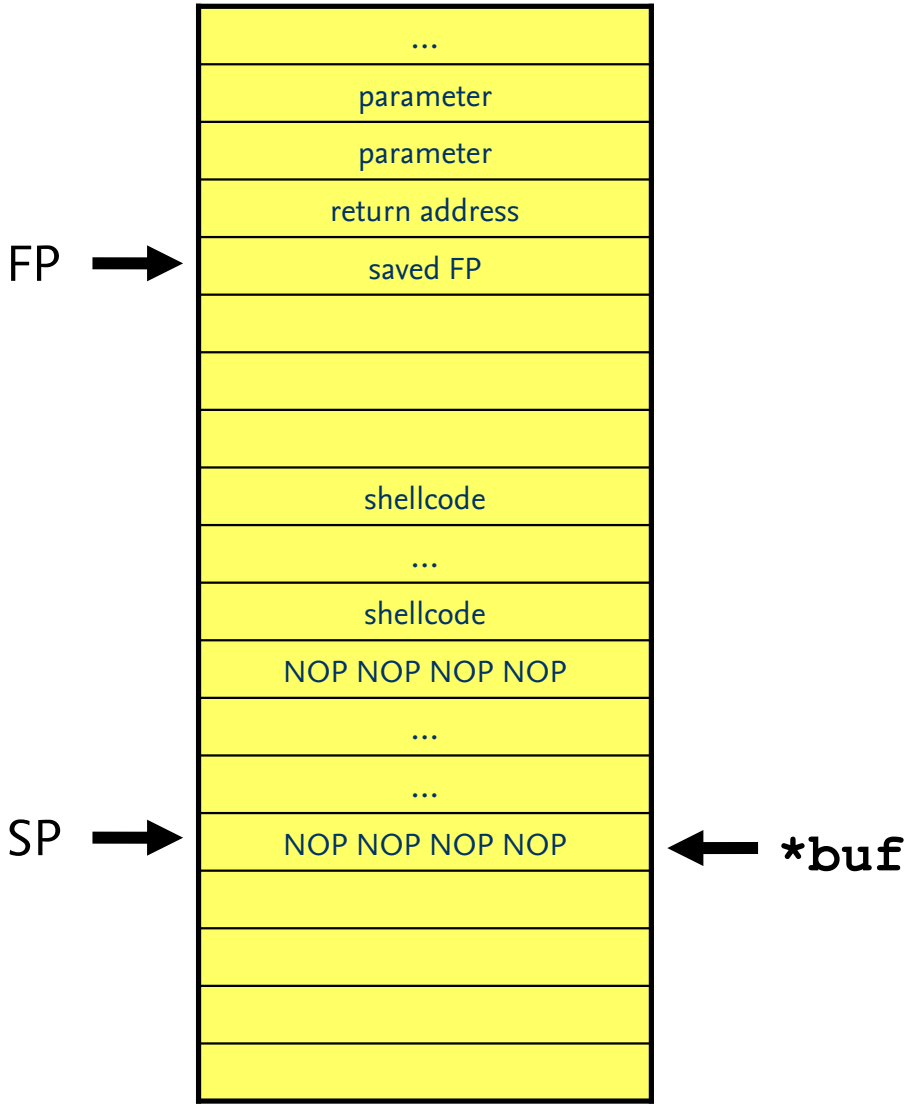
Buffer Overflow on the stack




```
int something(int,int){
  char buf[512];
  ...
  strcpy(buf,argc[1]);
}
```

- argc[1] contains the constructed malicious data
- The data starts with the NOPs
- In the middle follows the Shellcode

Buffer Overflow on the stack




```
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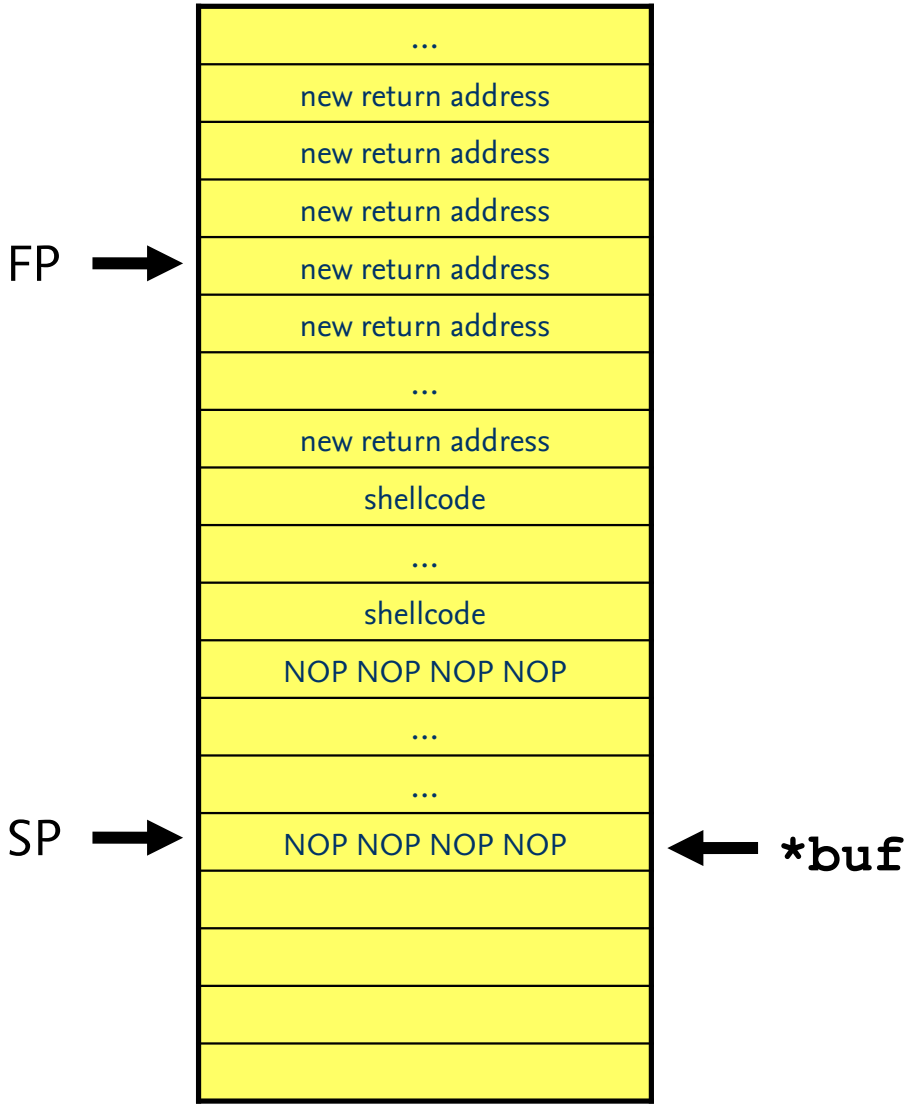
   char buf[512];
   ...
   strcpy(buf,argc[1]);
}
```

- argc[1] contains the constructed malicious data
- The data starts with the NOPs
- In the middle follows the Shellcode
- The remaining buffer is filled with the (guessed) malicious return address

Buffer Overflow on the stack

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   char buf[512];
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   strcpy(buf,argc[1]);
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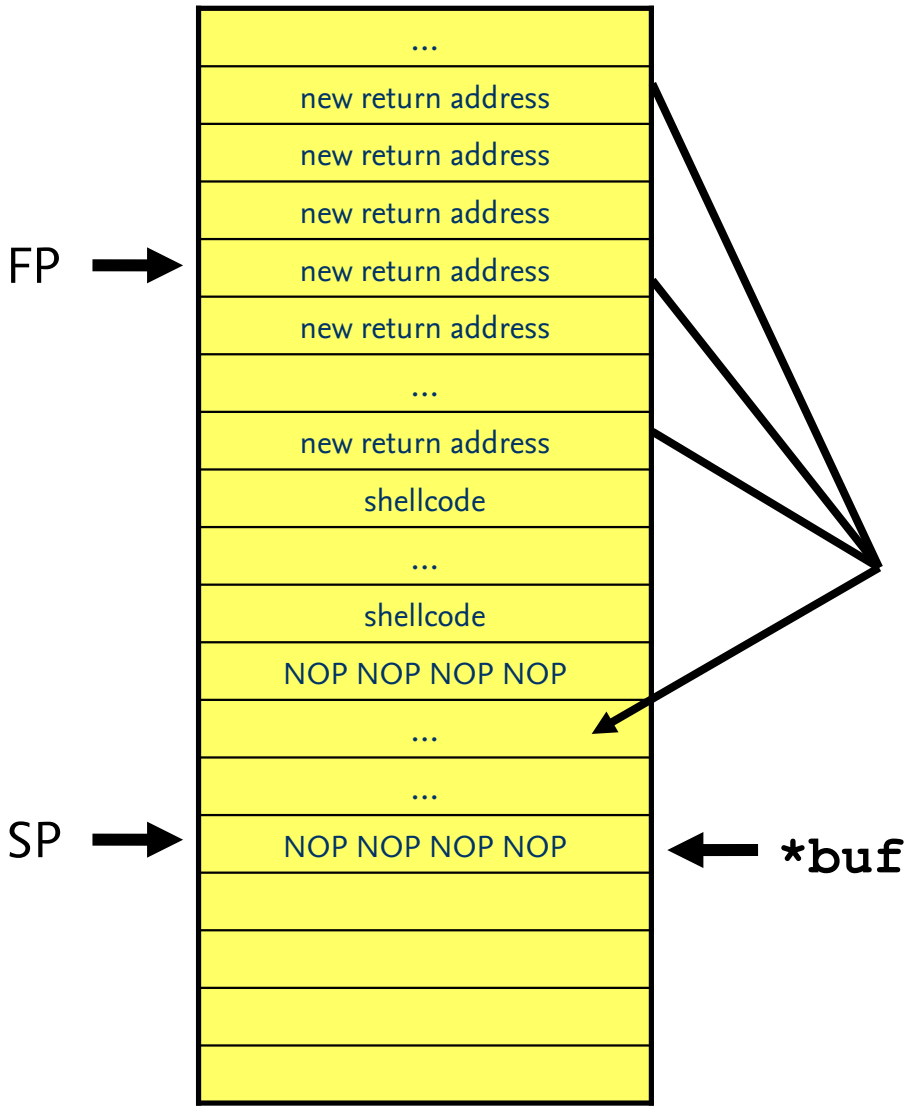
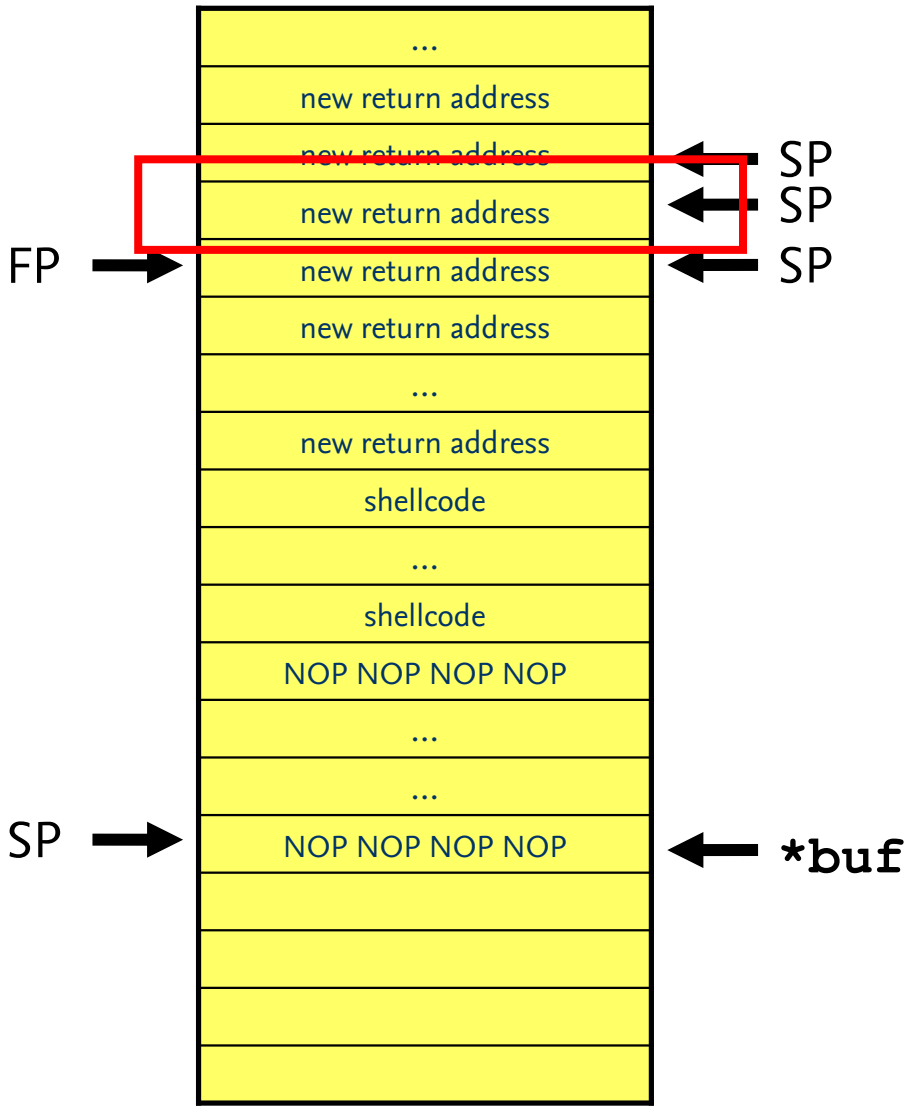



```
int something(int,int){
   char buf[512];
   ...
   strcpy(buf,argc[1]);
}
```

- The new return address points to a location in the buffer
- At least if the guessed location of the stack was correct

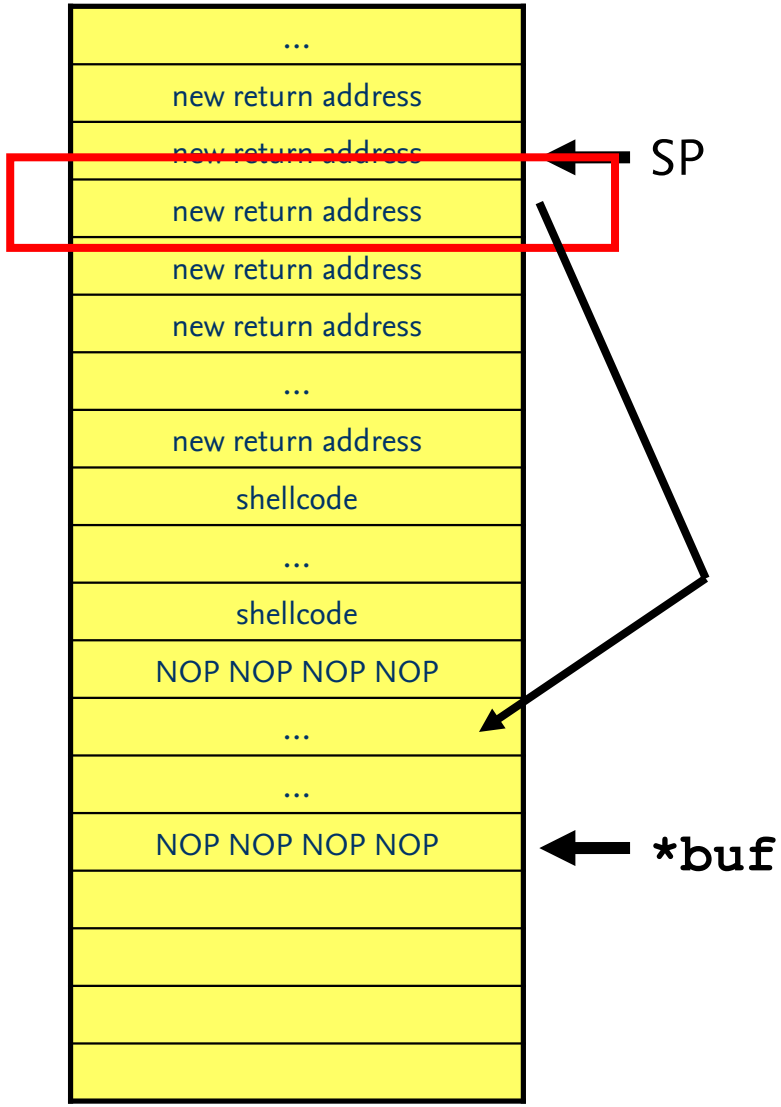
Buffer Overflow on the stack




```
int something(int,int){
  char buf[512];
  ...
  strcpy(buf,argc[1]);
  return(0);
}
```

- The Stack Pointer is moved to the Frame Pointer
- The old Frame Pointer is popped from the stack
- The old value was overwritten, the location of the FP is therefore meaningless
- The return address is popped from the stack into the Instruction Register (IR)

Buffer Overflow on the stack




```
int something(int,int){
  char buf[512];
  ...
  strcpy(buf,argc[1]);
  return(0);
}
```

- The new return address points to a location in the buffer
- The program executes the NOPs and the following shellcode

Exploiting a Buffer Overflow (II)

Small errors can have a profound effect

- "Of by One" errors
  - The buffer size is miscalculated by one
  - Example: `a[9]` has 10 elements
  - Result: possible manipulation of the frame pointer

Buffer Overflows on the heap are as dangerous as on the stack

- Function pointer manipulation
- Overwriting of C++ virtual pointer tables
- Exploitation of heap management functions
  - Highly dependent on the operating system
  - Sometimes in connection with further programming mistakes like for example "Double Free Errors"

Causes

Causes for Buffer Overflows (I)

- **Library functions, that don't check buffer boundaries:**
  `strcpy(), strcat(), sprintf(),…`
- **Library functions, that rely on the programmer to pass correct buffer boundaries:**

```
char buf1[4];
char buf2[8];
buf2 = "1234567";
strncpy(buf1,buf2,8);
```

- **Careless pointer arithmetic:**

```
while(<some condition>){
    *buf1++ = *buf2++
}
```

  (cause of the Blaster Worm)
- **Careless type casting of variables**

Programming Techniques

- **Don't use C, C++ or Objective C**

- Use "safer" library replacements
  - String libraries:
    - strsafe (Microsoft)
    - strlcat/strlcpy (Unix)
  - Integer:
    - SafeInt (Microsoft)
- Replace C Strings with C++ Strings
- C++: Replace static arrays with STL containers
- Avoid pointer arithmetic if possible

Thank you
for your attention

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